WEJ LAB

Advanced Topics in Regression

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Introduction to Regression

We often want to know how two or more variables are related.

- Example: which baseball player statistics predict team success, to build a team on a limited budget?
- First, the core concepts of regression.

Introduction to Regression

Regression estimates the relationship between:

- a **dependent variable**: outcome, response, or *label* in ML
- one or more **independent variables**: predictors, covariates, or *features* in ML

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Introduction to Regression

The most common form is **linear regression**: find the line (or linear combination) that best fits the data under a chosen criterion.

Introduction to Regression

In math class, a line is (slope m, intercept b):

$$y = mx + b.$$

In statistics, we write:

$$y_i = \beta_0 + x_i\beta_1,$$

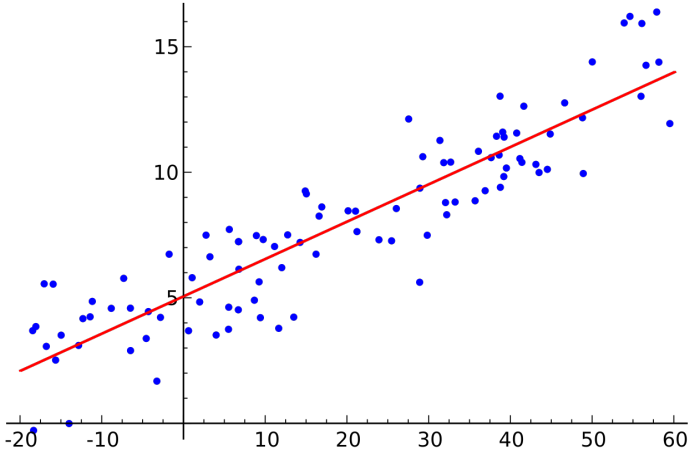
with slope β_1, intercept β_0, and observed data points (x_i, y_i).

Introduction to Regression

- In math, lines fit the points *exactly*
- In statistics, points *do not* fall exactly on the line
- We find the **best fit** line to estimate the unknown slope and intercept: a **linear regression model**

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Introduction to Regression



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Introduction to Regression

Formally, we observe n data pairs

$$(x_i, y_i), i = 1, ..., n,$$

and model

$$y_i = \beta_0 + x_i\beta_1 + \epsilon_i.$$

- ϵ_i are the **errors** (deviations from the line)
- β_0 and β_1 are unknown and must be *estimated* from the data

Introduction to Regression

Goal: find estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ that "best fit" the data.

Least squares: minimize the sum of squared residuals

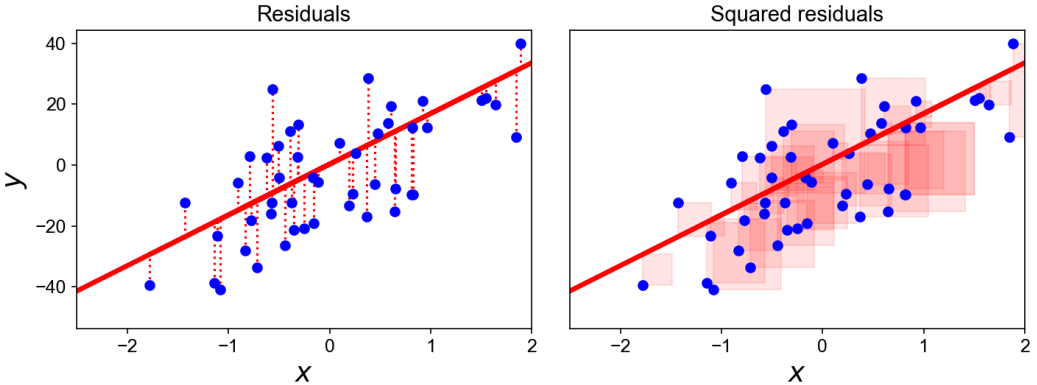
$$\min_{\beta_0,\beta_1} Q(\beta_0,\beta_1) = \inf_{\beta_0,\beta_1} \left\{ \sum_{i=1}^{n} (y_i - \beta_0 - x_i\beta_1)^2 \right\} = \inf_{\beta_0,\beta_1} \sum_{i=1}^{n} \epsilon_i^2.$$

Introduction to Regression

The least squares estimator (LSE) is the most common choice because of its properties:

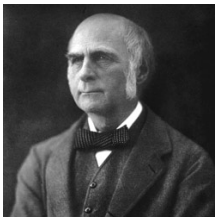
- It is the *best linear unbiased estimator* (Gauss-Markov)
- No other linear unbiased estimator has smaller sampling variance for $\hat{\beta}_0$ and $\hat{\beta}_1$

Introduction to regression



Case study: is height hereditary?

Francis Galton[1] studied heredity using family height data. Along the way, he developed **correlation** and **regression**.



[1] https://en.wikipedia.org/wiki/Francis_Galton

Case study: is height hereditary?

Galton's question: how well can we predict a child's height from the parents' heights? His answer: **regression**!

Historical note: Galton also promoted eugenics, a scientifically flawed movement with horrific consequences: https://pged.org/history-eugenics-and-genetics/.

Case study: is height hereditary?

Galton's data are in `HistData`. Keep each father and one randomly selected son:

```
library(tidyverse)
library(HistData)
data("GaltonFamilies")

set.seed(1983)
galton_heights <- GaltonFamilies %>%
  filter(gender == "male") %>%
  group_by(family) %>%
  sample_n(1) %>%
  ungroup() %>%
  select(father, childHeight) %>%
  rename(son = childHeight)
```

Case study: is height hereditary?

Both distributions are approximately normal, so summarize with means and SDs:

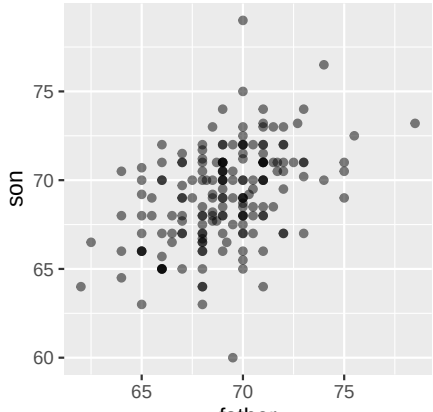
```
galton_heights %>%
  summarize(mean(father), sd(father), mean(son), sd(son))
```

```
## # A tibble: 1 x 4
##   `mean(father)` `sd(father)` `mean(son)` `sd(son)`
##            <dbl>        <dbl>       <dbl>     <dbl>
## 1           69.1         2.55        69.2      2.71
```

But this misses the key feature: taller fathers have taller sons.

Case study: is height hereditary?

```
galton_heights %>% ggplot(aes(father, son)) +
  geom_point(alpha = 0.5)
```



Case study: is height hereditary?

Correlation between father and son heights:

```
galton_heights %>%
    summarize(r = cor(father, son)) %>%
    pull(r)
```

```
## [1] 0.4334102
```

Case study: is height hereditary?

Fit the least squares model with `lm`:

$$y_i = \beta_0 + x_i\beta_1 + \varepsilon_i$$

(y_i = son's height, x_i = father's height)

```
fit <- lm(son ~ father, data = galton_heights)
fit$coef
```

```
## (Intercept)      father
##   37.287605    0.461392
```

Case study: is height hereditary?

summary gives more detail about the fit:

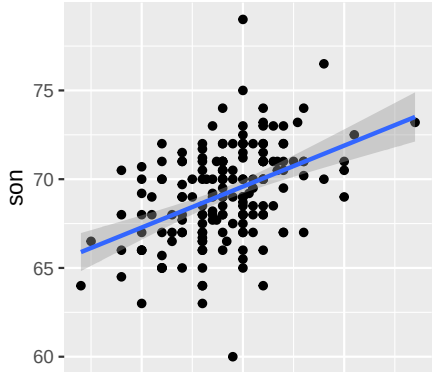
```
summary(fit)
```

```
##
## Call:
## lm(formula = son ~ father, data = galton_heights)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.3543 -1.5657 -0.0078  1.7263  9.4150
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 37.28761    4.98618   7.478 3.37e-12 ***
## father       0.46139    0.07211   6.398 1.36e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.45 on 177 degrees of freedom
## Multiple R-squared:  0.1878, Adjusted R-squared:  0.1833
```

Case study: is height hereditary?

Plot $\hat{Y}$ with confidence intervals in ggplot2:

```
galton_heights %>% ggplot(aes(father, son)) +
  geom_point() + geom_smooth(method = "lm")
```



Case study: is height hereditary?

`predict` returns predictions from an `lm` object, plus standard errors if requested:

```
fit <- galton_heights %>% lm(son ~ father, data = .)
y_hat <- fit %>% predict(se.fit = TRUE)
names(y_hat)
```

```
## [1] "fit"            "se.fit"         "df"             "residual.scale"
```

Introduction to Regression

Extend to p predictors:

$$y_i = x_{i1}\beta_1 + \ldots + x_{ip}\beta_p + \epsilon_i = \sum_{j=1}^{p} x_{ij}\beta_j + \epsilon_i.$$

To include an intercept, make one predictor constant ($x_{ij} = 1$ for all i), so $x_{ij}\beta_j = \beta_0$.

Introduction to Regression

Multiple regression on the full dataset:

```
fit2 <- GaltonFamilies %>%
  lm(childHeight ~ mother + father + gender + children, data = .)
fit2 %>% summary()
```

```
##
## Call:
## lm(formula = childHeight ~ mother + father + gender + children,
##     data = .)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.4759 -1.4743  0.0906  1.4789  9.1734
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 17.43103    2.77407   6.284 5.08e-10 ***
## mother       0.31619    0.03098  10.207  < 2e-16 ***
## father       0.38521    0.02898  13.292  < 2e-16 ***
## gendermale   5.19852    0.14197  36.617  < 2e-16 ***
## children    -0.04573    0.02631  -1.738   0.0825 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.162 on 929 degrees of freedom
## Multiple R-squared:  0.6366, Adjusted R-squared:  0.635
```

Introduction to Regression

Interpreting the model:

- Intercept: $\beta_0 = 17.4$ inches (p = 5.08e−10)
- Each inch of mother's height: $+0.32$ inches (p < 2e−16)
- Each inch of father's height: $+0.39$ inches (p < 2e−16)
- Male vs. female (reference): $+5.2$ inches (p < 2e−16). *R uses the first factor level (alphabetical) as the reference.*
- Number of children: not significant (p = 0.0825)
- $R^2 = 0.637$ (adjusted 0.635): `father`, `mother`, and `gender` are moderately strong predictors

Introduction to Regression

In matrix form, with outcome $\mathbf{y} = (y_1, \ldots, y_n)$, predictor matrix $\mathbf{X}$, and coefficients β $(p \times 1)$:

$$\mathbf{y} = X\beta + \epsilon, \quad \epsilon \sim N(\mathbf{0}, \sigma^2 I_n).$$

Introduction to Regression

Least squares then minimizes:

$$\hat{\beta} = \inf_{\beta} \left\{ \sum_{i=1}^{n} (y_i - \sum_{j=1}^{p} x_{ij}\beta_j)^2 \right\} = \inf_{\beta} \{(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)\}$$

Diversion: Maximum Likelihood for a Normal mean

Let $x_1, \ldots, x_N$ be independent Normal(μ, σ^2), with σ^2 known. The likelihood is

$$L(\mu|\mathbf{x}) = \left\{\frac{1}{2\pi\sigma^2}\right\}^{N/2} exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^{N}(x_i - \mu)^2\right\}$$

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Maximizing $L(\mu|\mathbf{x})$ = minimizing $\sum_{i=1}^{N}(x_i - \mu)^2$ over μ.

Diversion: Maximum Likelihood for a Normal mean

Note the following:

$$\sum_{i=1}^{N}(x_i - \mu)^2 = \sum_{i=1}^{N}(x_i^2 - 2\mu x_i + \mu^2)$$

Diversion: Maximum Likelihood for a Normal mean

Note the following:

$$\begin{aligned}\sum_{i=1}^{N}(x_i-\mu)^2 &= \sum_{i=1}^{N}(x_i^2-2\mu x_i+\mu^2)\\ &= \sum_{i=1}^{N}x_i^2-2\mu\sum_{i=1}^{N}x_i+N\mu^2\end{aligned}$$

Diversion: Maximum Likelihood for a Normal mean

Note the following:

$$\begin{aligned}\sum_{i=1}^{N}(x_i-\mu)^2 &= \sum_{i=1}^{N}(x_i^2-2\mu x_i+\mu^2)\\ &= \sum_{i=1}^{N}x_i^2-2\mu\sum_{i=1}^{N}x_i+N\mu^2\end{aligned}$$

Diversion: Maximum Likelihood for a Normal mean

therefore:

$$\frac{\partial}{\partial \mu} = -2 \sum x_i + 2N\hat{\mu} \stackrel{set}{=} 0$$

Diversion: Maximum Likelihood for a Normal mean

therefore:

$$\frac{\partial}{\partial \mu} = -2 \sum x_i + 2N\hat{\mu} \stackrel{set}{=} 0$$

$$\Rightarrow N\hat{\mu} = \sum x_i$$

Diversion: Maximum Likelihood for a Normal mean

therefore:

$$\begin{aligned}\frac{\partial}{\partial \mu} &= -2\sum x_i + 2N\hat{\mu} \overset{set}{=} 0 \\ &\Rightarrow N\hat{\mu} = \sum x_i \\ &\Rightarrow \hat{\mu} = \frac{\sum x_i}{N}\end{aligned}$$

Diversion: Maximum Likelihood for Regression

Recall the regression model:

$$\mathbf{y} = X\beta + \epsilon, \quad \epsilon \sim N(\mathbf{0}, \sigma^2 I_N).$$

Diversion: Maximum Likelihood for Regression

Extending the Normal mean MLE:

$$\sum_{i=1}^{N}(y_i - x_{i1}\beta_1 - \ldots - x_{ip}\beta_p)^2 = (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta),$$

Diversion: Maximum Likelihood for Regression

Extending the Normal mean MLE:

$$\sum_{i=1}^{N}(y_i - x_{i1}\beta_1 - \ldots - x_{ip}\beta_p)^2 = (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta),$$

so

$$L(\beta|\mathbf{X}, \mathbf{y}) = \left\{\frac{1}{2\pi\sigma^2}\right\}^{N/2} exp\left\{-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)\right\}.$$

Maximizing the likelihood = minimizing the sum of squared errors!

Diversion: Maximum Likelihood for Regression

Note

$$(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) = \mathbf{y}'\mathbf{y} - 2\beta'\mathbf{X}'\mathbf{y} + \beta'\mathbf{X}'\mathbf{X}\beta,$$

Diversion: Maximum Likelihood for Regression

Note

$$(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) = \mathbf{y}'\mathbf{y} - 2\beta'\mathbf{X}'\mathbf{y} + \beta'\mathbf{X}'\mathbf{X}\beta,$$

and therefore

$$\frac{\partial}{\partial \beta} = -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\hat{\beta} \stackrel{set}{=} 0$$

Diversion: Maximum Likelihood for Regression

Note

$$(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) = \mathbf{y}'\mathbf{y} - 2\beta'\mathbf{X}'\mathbf{y} + \beta'\mathbf{X}'\mathbf{X}\beta,$$

and therefore

$$\begin{aligned}\frac{\partial}{\partial\beta} &= -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\hat{\beta} \stackrel{set}{=} 0 \\ &\Rightarrow \mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y}\end{aligned}$$

Diversion: Maximum Likelihood for Regression

Note

$$(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) = \mathbf{y}'\mathbf{y} - 2\beta'\mathbf{X}'\mathbf{y} + \beta'\mathbf{X}'\mathbf{X}\beta,$$

and therefore

$$\begin{aligned}
\frac{\partial}{\partial \beta} &= -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\hat{\beta} \stackrel{set}{=} 0 \\
&\Rightarrow \mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y} \\
&\Rightarrow (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}
\end{aligned}$$

Diversion: Maximum Likelihood for Regression

Note

$$(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) = \mathbf{y}'\mathbf{y} - 2\beta'\mathbf{X}'\mathbf{y} + \beta'\mathbf{X}'\mathbf{X}\beta,$$

and therefore

$$\begin{aligned}
\frac{\partial}{\partial \beta} &= -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\hat{\beta} \stackrel{set}{=} 0 \\
&\Rightarrow \mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y} \\
&\Rightarrow (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \\
&\Rightarrow \hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}
\end{aligned}$$

Diversion: Maximum Likelihood for Regression

Now note:

$$E[\hat{\beta}] = E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}]$$

Diversion: Maximum Likelihood for Regression

Now note:

$$
\begin{aligned}
E[\hat{\beta}] &= E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}] \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E[\mathbf{y}]
\end{aligned}
$$

Diversion: Maximum Likelihood for Regression

Now note:

$$
\begin{aligned}
E[\hat{\beta}] &= E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}] \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E[\mathbf{y}] \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta
\end{aligned}
$$

Diversion: Maximum Likelihood for Regression

Now note:

$$
\begin{aligned}
E[\hat{\beta}] &= E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}] \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E[\mathbf{y}] \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta \\
&= \beta
\end{aligned}
$$

Diversion: Maximum Likelihood for Regression

And, remembering that

$$Var[\mathbf{Ay}] = \mathbf{A}\{Var[\mathbf{y}]\}\mathbf{A}',$$

so therefore,

$$Var[\hat{\beta}] = Var[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}]$$

Diversion: Maximum Likelihood for Regression

And, remembering that

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so therefore,

$$\begin{aligned} Var[\hat{\beta}] &= Var[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}] \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\{Var[\mathbf{y}]\}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \end{aligned}$$

Diversion: Maximum Likelihood for Regression

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Diversion: Maximum Likelihood for Regression

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Diversion: Maximum Likelihood for Regression

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so therefore,

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Diversion: Maximum Likelihood for Regression

Test $H_0 : \beta_j = 0$ vs. $H_a : \beta_j \neq 0$ with:

$$t_{\beta_j} = \frac{\hat{\beta}_j}{\hat{\sigma}\sqrt{v_{jj}}},$$

where v_{jj} is the jth diagonal of $\mathbf{V} = (\mathbf{X}'\mathbf{X})^{-1}$. Under H_0, $t_{\beta_j} \sim t_{N-p-1}$.

Case study: is height hereditary?

Using our estimator:

$$\widehat{\beta} = (X'X)^{-1}X'\mathbf{y}$$

```
X <- cbind(1,galton_heights$father)
y <- galton_heights$son
beta_hat <- solve(t(X)%*%X)%*%t(X)%*%y
beta_hat
```

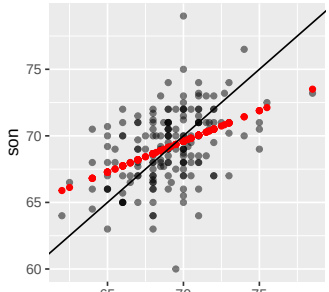
```
##             [,1]
## [1,] 37.287605
## [2,]  0.461392
```

Case study: is height hereditary?

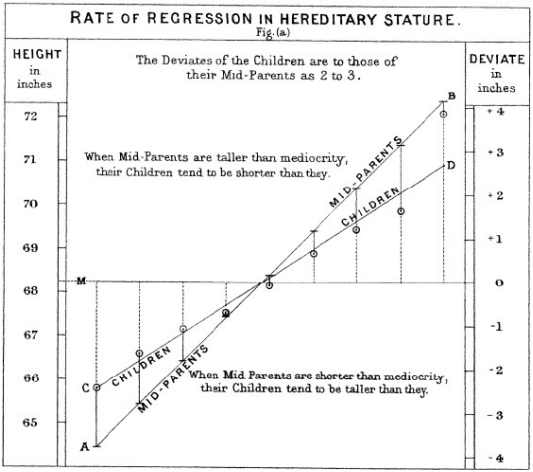
Predicted values: $\hat{y} = X\hat{\beta}$

```
y_hat <- X %*% beta_hat

galton_heights %>% ggplot(aes(father, son)) +
  geom_point(alpha = 0.5) +
  geom_point(aes(y = y_hat), col = "red") +
  geom_abline(slope = 1, intercept = 0)
```



Case study: is height hereditary?



Case study: is height hereditary?

Standard error for hypothesis testing:

$$SE(\hat{\beta}_i) = \hat{\sigma}\sqrt{v_{ii}},$$

where v_{ii} is the ith diagonal element of $(X'X)^{-1}$, and

$$\hat{\sigma}^2 = \frac{1}{N-p-1}(\mathbf{y} - X\widehat{\beta})'(\mathbf{y} - X\widehat{\beta})$$

```
N <- length(y)
p <- length(beta_hat)
sigma2_hat <- 1 / (N - p - 1) *
  t(y - X %*% beta_hat) %*% (y - X %*% beta_hat)
V <- solve(t(X) %*% X)
Z2 <- beta_hat[2] / sqrt(sigma2_hat * V[2, 2])
Z2
```

The Regression Fallacy

The *sophomore slump* (Wikipedia):

> *A second, or sophomore, effort fails to live up to the standards of the first: students, athletes (second season), bands (second album), TV shows, and movie sequels.*

The Regression Fallacy

- MLB's Rookie of the Year (ROY) goes to the best first-year player
- ROY winners often do worse in year two: the "sophomore slump"
- Fox Sports[2]: "Will MLB's tremendous rookie class of 2015 suffer a sophomore slump?"

[2] http://www.foxsports.com/mlb/story/kris-bryant-carlos-correa-rookies-of-year-award-matt-duffy-francisco-lindor-kang-sano-120715

The Regression Fallacy

Is it real? Batting averages for the top ROYs do drop:

nameFirst	nameLast	rookie_year	rookie	sophomore
Willie	McCovey	1959	0.3541667	0.2384615
Ichiro	Suzuki	2001	0.3497110	0.3214838
Al	Bumbry	1973	0.3370787	0.2333333
Fred	Lynn	1975	0.3314394	0.3136095
Albert	Pujols	2001	0.3288136	0.3135593

The Regression Fallacy

Proportion of ROYs with a lower sophomore average: 0.7053571.

Jitters or jinx? Look at all players with 130+ at bats (the ROY minimum) in both 2013 and 2014.

The Regression Fallacy

Same pattern: most top performers drop.

nameFirst	nameLast	2013	2014
Miguel	Cabrera	0.3477477	0.3126023
Hanley	Ramirez	0.3453947	0.2828508
Michael	Cuddyer	0.3312883	0.3315789
Scooter	Gennett	0.3239437	0.2886364
Joe	Mauer	0.3235955	0.2769231

But these are not rookies!

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The Regression Fallacy

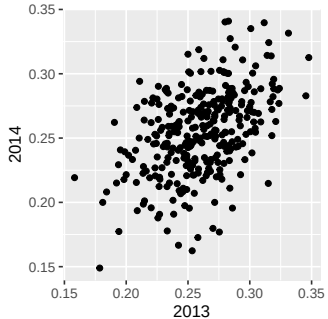
Now the worst performers of 2013:

nameFirst	nameLast	2013	2014
Danny	Espinosa	0.1582278	0.2192192
Dan	Uggla	0.1785714	0.1489362
Jeff	Mathis	0.1810345	0.2000000
B. J.	Upton	0.1841432	0.2080925
Adam	Rosales	0.1904762	0.2621951

Their batting averages mostly go up!

The Regression Fallacy

A reverse slump? No. There is no sophomore slump: year-to-year correlation is high, but not perfect.



The Regression Fallacy

Correlation is 0.460254 and the data look bivariate normal, so we predict 2014 average Y from 2013 average X with:

$$\frac{Y - .255}{.032} = 0.46 \left(\frac{X - .261}{.023} \right)$$

The Regression Fallacy

- Correlation < 1, so top 2013 performers are expected to do a bit worse in 2014
- Not a jinx, just chance
- ROYs are selected from the top of X, so Y will **regress to the mean**

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Session Info

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.6.2
##
## Matrix products: default
## BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/libBLAS.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  LAPACK version 3.12.1
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/Denver
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] Lahman_14.0-0   HistData_1.0.0  lubridate_1.9.5 forcats_1.0.1
##  [5] stringr_1.6.0   dplyr_1.2.1     purrr_1.2.2     readr_2.2.0
##  [9] tidyr_1.3.2     tibble_3.3.1    ggplot2_4.0.3   tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] generics_0.1.4     xml2_1.6.0         stringi_1.8.9      lattice_0.22-9
##  [5] hms_1.1.4          digest_0.6.39      magrittr_2.0.5     evaluate_1.0.5
##  [9] grid_4.5.2         timechange_0.4.0   RColorBrewer_1.1-3 fastmap_1.2.0
```

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